

1. Introduction

To convince the reader that stochastic differential equations is an important subject let us mention some situations where such equations appear and can be used:

1.1 Stochastic Analogs of Classical Differential Equations

If we allow for some randomness in some of the coefficients of a differential equation we often obtain a more realistic mathematical model of the situation.

Problem 1. Consider the simple population growth model

$$\frac{dN}{dt} = a(t)N(t), \quad N(0) = N_0 \text{ (constant)} \quad (1.1.1)$$

where $N(t)$ is the size of the population at time t , and $a(t)$ is the relative rate of growth at time t . It might happen that $a(t)$ is not completely known, but subject to some random environmental effects, so that we have

$$a(t) = r(t) + \text{“noise”} ,$$

where we do not know the exact behaviour of the noise term, only its probability distribution. The function $r(t)$ is assumed to be nonrandom. How do we solve (1.1.1) in this case?

Problem 2. The charge $Q(t)$ at time t at a fixed point in an electric circuit satisfies the differential equation

$$L \cdot Q''(t) + R \cdot Q'(t) + \frac{1}{C} \cdot Q(t) = F(t), \quad Q(0) = Q_0, \quad Q'(0) = I_0 \quad (1.1.2)$$

where L is inductance, R is resistance, C is capacitance and $F(t)$ the potential source at time t .

Again we may have a situation where some of the coefficients, say $F(t)$, are not deterministic but of the form

$$F(t) = G(t) + \text{“noise”} . \quad (1.1.3)$$

How do we solve (1.1.2) in this case?

More generally, the equation we obtain by allowing randomness in the coefficients of a differential equation is called a *stochastic differential equation*. This will be made more precise later. It is clear that any solution of a stochastic differential equation must involve some randomness, i.e. we can only hope to be able to say something about the probability distributions of the solutions.

1.2 Filtering Problems

Problem 3. Suppose that we, in order to improve our knowledge about the solution, say of Problem 2, perform observations $Z(s)$ of $Q(s)$ at times $s \leq t$. However, due to inaccuracies in our measurements we do not really measure $Q(s)$ but a disturbed version of it:

$$Z(s) = Q(s) + \text{“noise”} . \quad (1.2.1)$$

So in this case there are two sources of noise, the second coming from the error of measurement.

The *filtering problem* is: What is the best estimate of $Q(t)$ satisfying (1.1.2), based on the observations Z_s in (1.2.1), where $s \leq t$? Intuitively, the problem is to “filter” the noise away from the observations in an optimal way.

In 1960 Kalman and in 1961 Kalman and Bucy proved what is now known as the Kalman-Bucy filter. Basically the filter gives a procedure for estimating the state of a system which satisfies a “noisy” linear differential equation, based on a series of “noisy” observations.

Almost immediately the discovery found applications in aerospace engineering (Ranger, Mariner, Apollo etc.) and it now has a broad range of applications.

Thus the Kalman-Bucy filter is an example of a recent mathematical discovery which has already proved to be useful – it is not just “potentially” useful.

It is also a counterexample to the assertion that “applied mathematics is bad mathematics” and to the assertion that “the only really useful mathematics is the elementary mathematics”. For the Kalman-Bucy filter – as the whole subject of stochastic differential equations – involves advanced, interesting and first class mathematics.

1.3 Stochastic Approach to Deterministic Boundary Value Problems

Problem 4. The most celebrated example is the stochastic solution of the *Dirichlet problem*:

Given a (reasonable) domain U in $\mathbf{R}^n$ and a continuous function f on the boundary of U , ∂U . Find a function $\tilde{f}$ continuous on the closure $\overline{U}$ of U such that

- (i) $\tilde{f} = f$ on ∂U
- (ii) $\tilde{f}$ is harmonic in U , i.e.

$$\Delta \tilde{f} := \sum_{i=1}^n \frac{\partial^2 \tilde{f}}{\partial x_i^2} = 0 \quad \text{in } U.$$

In 1944 Kakutani proved that the solution could be expressed in terms of *Brownian motion* (which will be constructed in Chapter 2): $\tilde{f}(x)$ is the expected value of f at the first exit point from U of the Brownian motion starting at $x \in U$.

It turned out that this was just the tip of an iceberg: For a large class of semielliptic second order partial differential equations the corresponding Dirichlet boundary value problem can be solved using a stochastic process which is a solution of an associated stochastic differential equation.

1.4 Optimal Stopping

Problem 5. Suppose a person has an asset or resource (e.g. a house, stocks, oil...) that she is planning to sell. The price X_t at time t of her asset on the open market varies according to a stochastic differential equation of the same type as in Problem 1:

$$\frac{dX_t}{dt} = rX_t + \alpha X_t \cdot \text{“noise”}$$

where r, α are known constants. The discount rate is a known constant ρ . At what time should she decide to sell?

We assume that she knows the behaviour of X_s up to the present time t , but because of the noise in the system she can of course never be sure at the time of the sale if her choice of time will turn out to be the best. So what we are searching for is a stopping strategy that gives the best result in the long run, i.e. maximizes the *expected* profit when the inflation is taken into account.

This is an *optimal stopping problem*. It turns out that the solution can be expressed in terms of the solution of a corresponding boundary value problem (Problem 4), except that the boundary is unknown (free) as well and this is compensated by a double set of boundary conditions. It can also be expressed in terms of a set of *variational inequalities*.

1.5 Stochastic Control

Problem 6 (An optimal portfolio problem).

Suppose that a person has two possible investments:

- (i) A *risky* investment (e.g. a stock), where the price $p_1(t)$ per unit at time t satisfies a stochastic differential equation of the type discussed in Problem 1:

$$\frac{dp_1}{dt} = (a + \alpha \cdot \text{"noise"})p_1 \quad (1.5.1)$$

where $a > 0$ and $\alpha \in \mathbf{R}$ are constants

- (ii) A *safe* investment (e.g. a bond), where the price $p_2(t)$ per unit at time t grows exponentially:

$$\frac{dp_2}{dt} = bp_2 \quad (1.5.2)$$

where b is a constant, $0 < b < a$.

At each instant t the person can choose how large portion (fraction) u_t of his fortune X_t he wants to place in the risky investment, thereby placing $(1 - u_t)X_t$ in the safe investment. Given a utility function U and a terminal time T the problem is to find the optimal *portfolio* $u_t \in [0, 1]$ i.e. find the investment distribution u_t ; $0 \leq t \leq T$ which maximizes the expected utility of the corresponding terminal fortune $X_T^{(u)}$:

$$\max_{0 \leq u_t \leq 1} \left\{ E \left[U(X_T^{(u)}) \right] \right\} \quad (1.5.3)$$

1.6 Mathematical Finance

Problem 7 (Pricing of options).

Suppose that at time $t = 0$ the person in Problem 6 is offered the *right* (but without obligation) to buy one unit of the risky asset at a specified price K and at a specified future time $t = T$. Such a right is called a *European call option*. How much should the person be willing to pay for such an option? This problem was solved when Fischer Black and Myron Scholes (1973) used stochastic analysis and an equilibrium argument to compute a theoretical value for the price, the now famous *Black and Scholes option price formula*. This theoretical value agreed well with the prices that had already been established as an equilibrium price on the free market. Thus it represented a triumph for mathematical modelling in finance. It has become an indispensable tool in the trading of options and other financial derivatives. In 1997 Myron Scholes and Robert Merton were awarded the Nobel Prize

in Economics for their work related to this formula. (Fischer Black died in 1995.)

We will return to these problems in later chapters, after having developed the necessary mathematical machinery. We solve Problem 1 and Problem 2 in Chapter 5. Problems involving filtering (Problem 3) are treated in Chapter 6, the generalized Dirichlet problem (Problem 4) in Chapter 9. Problem 5 is solved in Chapter 10 while stochastic control problems (Problem 6) are discussed in Chapter 11. Finally we discuss applications to mathematical finance in Chapter 12.

2. Some Mathematical Preliminaries

2.1 Probability Spaces, Random Variables and Stochastic Processes

Having stated the problems we would like to solve, we now proceed to find reasonable mathematical notions corresponding to the quantities mentioned and mathematical models for the problems. In short, here is a first list of the notions that need a mathematical interpretation:

- (1) A random quantity
- (2) Independence
- (3) Parametrized (discrete or continuous) families of random quantities
- (4) What is meant by a “best” estimate in the filtering problem (Problem 3)
- (5) What is meant by an estimate “based on” some observations (Problem 3)?
- (6) What is the mathematical interpretation of the “noise” terms?
- (7) What is the mathematical interpretation of the stochastic differential equations?

In this chapter we will discuss (1)–(3) briefly. In the next chapter we will consider (6), which leads to the notion of an Itô stochastic integral (7). In Chapter 6 we will consider (4)–(5).

The mathematical model for a random quantity is a *random variable*. Before we define this, we recall some concepts from general probability theory. The reader is referred to e.g. Williams (1991) for more information.

Definition 2.1.1. *If Ω is a given set, then a σ -algebra $\mathcal{F}$ on Ω is a family $\mathcal{F}$ of subsets of Ω with the following properties:*

- (i) $\emptyset \in \mathcal{F}$
- (ii) $F \in \mathcal{F} \Rightarrow F^C \in \mathcal{F}$, where $F^C = \Omega \setminus F$ is the complement of F in Ω
- (iii) $A_1, A_2, \dots \in \mathcal{F} \Rightarrow A := \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$

The pair $(\Omega, \mathcal{F})$ is called a measurable space. A probability measure P on a measurable space $(\Omega, \mathcal{F})$ is a function $P: \mathcal{F} \rightarrow [0, 1]$ such that

- (a) $P(\emptyset) = 0, P(\Omega) = 1$

(b) if $A_1, A_2, \dots \in \mathcal{F}$ and $\{A_i\}_{i=1}^\infty$ is disjoint (i.e. $A_i \cap A_j = \emptyset$ if $i \neq j$) then

$$P\left(\bigcup_{i=1}^\infty A_i\right) = \sum_{i=1}^\infty P(A_i) .$$

The triple $(\Omega, \mathcal{F}, P)$ is called a probability space. It is called a complete probability space if $\mathcal{F}$ contains all subsets G of Ω with P -outer measure zero, i.e. with

$$P^*(G) := \inf\{P(F); F \in \mathcal{F}, G \subset F\} = 0 .$$

Any probability space can be made complete simply by adding to $\mathcal{F}$ all sets of outer measure 0 and by extending P accordingly.

The subsets F of Ω which belong to $\mathcal{F}$ are called $\mathcal{F}$ -measurable sets. In a probability context these sets are called *events* and we use the interpretation

$$P(F) = \text{“the probability that the event } F \text{ occurs”} .$$

In particular, if $P(F) = 1$ we say that “ F occurs with probability 1”, or “almost surely (a.s.)”.

Given any family $\mathcal{U}$ of subsets of Ω there is a smallest σ -algebra $\mathcal{H}_{\mathcal{U}}$ containing $\mathcal{U}$, namely

$$\mathcal{H}_{\mathcal{U}} = \bigcap \{\mathcal{H}; \mathcal{H} \text{ } \sigma\text{-algebra of } \Omega, \mathcal{U} \subset \mathcal{H}\} .$$

(See Exercise 2.3.)

We call $\mathcal{H}_{\mathcal{U}}$ the σ -algebra generated by $\mathcal{U}$.

For example, if $\mathcal{U}$ is the collection of all open subsets of a topological space Ω (e.g. $\Omega = \mathbf{R}^n$), then $\mathcal{B} = \mathcal{H}_{\mathcal{U}}$ is called the *Borel σ -algebra* on Ω and the elements $B \in \mathcal{B}$ are called *Borel sets*. $\mathcal{B}$ contains all open sets, all closed sets, all countable unions of closed sets, all countable intersections of such countable unions etc.

If $(\Omega, \mathcal{F}, P)$ is a given probability space, then a function $Y: \Omega \rightarrow \mathbf{R}^n$ is called $\mathcal{F}$ -measurable if

$$Y^{-1}(U) := \{\omega \in \Omega; Y(\omega) \in U\} \in \mathcal{F}$$

for all open sets $U \subset \mathbf{R}^n$ (or, equivalently, for all Borel sets $U \subset \mathbf{R}^n$).

If $X: \Omega \rightarrow \mathbf{R}^n$ is any function, then the σ -algebra $\mathcal{H}_X$ generated by X is the smallest σ -algebra on Ω containing all the sets

$$X^{-1}(U) ; \quad U \subset \mathbf{R}^n \text{ open} .$$

It is not hard to show that

$$\mathcal{H}_X = \{X^{-1}(B); B \in \mathcal{B}\} ,$$

where $\mathcal{B}$ is the Borel σ -algebra on $\mathbf{R}^n$. Clearly, X will then be $\mathcal{H}_X$ -measurable and $\mathcal{H}_X$ is the smallest σ -algebra with this property.

The following result is useful. It is a special case of a result sometimes called the *Doob-Dynkin lemma*. See e.g. M. M. Rao (1984), Prop. 3, p. 7.

Lemma 2.1.2. *If $X, Y: \Omega \rightarrow \mathbf{R}^n$ are two given functions, then Y is $\mathcal{H}_X$ -measurable if and only if there exists a Borel measurable function $g: \mathbf{R}^n \rightarrow \mathbf{R}^n$ such that*

$$Y = g(X) .$$

In the following we let $(\Omega, \mathcal{F}, P)$ denote a given complete probability space. A *random variable* X is an $\mathcal{F}$ -measurable function $X: \Omega \rightarrow \mathbf{R}^n$. Every random variable induces a probability measure μ_X on $\mathbf{R}^n$, defined by

$$\mu_X(B) = P(X^{-1}(B)) .$$

μ_X is called the *distribution of X* .

If $\int_{\Omega} |X(\omega)| dP(\omega) < \infty$ then the number

$$E[X] := \int_{\Omega} X(\omega) dP(\omega) = \int_{\mathbf{R}^n} x d\mu_X(x)$$

is called the *expectation of X* (w.r.t. P).

More generally, if $f: \mathbf{R}^n \rightarrow \mathbf{R}$ is Borel measurable and $\int_{\Omega} |f(X(\omega))| dP(\omega) < \infty$ then we have

$$E[f(X)] := \int_{\Omega} f(X(\omega)) dP(\omega) = \int_{\mathbf{R}^n} f(x) d\mu_X(x) .$$

The mathematical model for independence is the following:

Definition 2.1.3. *Two subsets $A, B \in \mathcal{F}$ are called independent if*

$$P(A \cap B) = P(A) \cdot P(B) .$$

A collection $\mathcal{A} = \{\mathcal{H}_i; i \in I\}$ of families $\mathcal{H}_i$ of measurable sets is independent if

$$P(H_{i_1} \cap \cdots \cap H_{i_k}) = P(H_{i_1}) \cdots P(H_{i_k})$$

for all choices of $H_{i_1} \in \mathcal{H}_{i_1}, \dots, H_{i_k} \in \mathcal{H}_{i_k}$ with different indices $i_1, \dots, i_k$.

A collection of random variables $\{X_i; i \in I\}$ is independent if the collection of generated σ -algebras $\mathcal{H}_{X_i}$ is independent.

If two random variables $X, Y: \Omega \rightarrow \mathbf{R}$ are independent then

$$E[XY] = E[X]E[Y] ,$$

provided that $E[|X|] < \infty$ and $E[|Y|] < \infty$. (See Exercise 2.5.)

Definition 2.1.4. *A stochastic process is a parametrized collection of random variables*

$$\{X_t\}_{t \in T}$$

defined on a probability space $(\Omega, \mathcal{F}, P)$ and assuming values in $\mathbf{R}^n$.

The parameter space T is usually (as in this book) the halflife $[0, \infty)$, but it may also be an interval $[a, b]$, the non-negative integers and even subsets of $\mathbf{R}^n$ for $n \geq 1$. Note that for each $t \in T$ fixed we have a random variable

$$\omega \rightarrow X_t(\omega) ; \quad \omega \in \Omega .$$

On the other hand, fixing $\omega \in \Omega$ we can consider the function

$$t \rightarrow X_t(\omega) ; \quad t \in T$$

which is called a *path* of X_t .

It may be useful for the intuition to think of t as “time” and each ω as an individual “particle” or “experiment”. With this picture $X_t(\omega)$ would represent the position (or result) at time t of the particle (experiment) ω . Sometimes it is convenient to write $X(t, \omega)$ instead of $X_t(\omega)$. Thus we may also regard the process as a function of two variables

$$(t, \omega) \rightarrow X(t, \omega)$$

from $T \times \Omega$ into $\mathbf{R}^n$. This is often a natural point of view in stochastic analysis, because (as we shall see) there it is crucial to have $X(t, \omega)$ jointly measurable in (t, ω) .

Finally we note that we may identify each ω with the function $t \rightarrow X_t(\omega)$ from T into $\mathbf{R}^n$. Thus we may regard Ω as a subset of the space $\tilde{\Omega} = (\mathbf{R}^n)^T$ of all functions from T into $\mathbf{R}^n$. Then the σ -algebra $\mathcal{F}$ will contain the σ -algebra $\mathcal{B}$ generated by sets of the form

$$\{\omega; \omega(t_1) \in F_1, \dots, \omega(t_k) \in F_k\} , \quad F_i \subset \mathbf{R}^n \text{ Borel sets}$$

($\mathcal{B}$ is the same as the Borel σ -algebra on $\tilde{\Omega}$ if $T = [0, \infty)$ and $\tilde{\Omega}$ is given the product topology). Therefore one may also adopt the point of view that a stochastic process is a *probability measure P on the measurable space $((\mathbf{R}^n)^T, \mathcal{B})$* .

The (*finite-dimensional*) *distributions* of the process $X = \{X_t\}_{t \in T}$ are the measures $\mu_{t_1, \dots, t_k}$ defined on $\mathbf{R}^{nk}$, $k = 1, 2, \dots$, by

$$\mu_{t_1, \dots, t_k}(F_1 \times F_2 \times \dots \times F_k) = P[X_{t_1} \in F_1, \dots, X_{t_k} \in F_k] ; \quad t_i \in T .$$

Here $F_1, \dots, F_k$ denote Borel sets in $\mathbf{R}^n$.

The family of all finite-dimensional distributions determine many (but not all) important properties of the process X .

Conversely, given a family $\{\nu_{t_1, \dots, t_k}; k \in \mathbf{N}, t_i \in T\}$ of probability measures on $\mathbf{R}^{nk}$ it is important to be able to construct a stochastic process $Y = \{Y_t\}_{t \in T}$ having $\nu_{t_1, \dots, t_k}$ as its finite-dimensional distributions. One of Kolmogorov’s famous theorems states that this can be done provided $\{\nu_{t_1, \dots, t_k}\}$ satisfies two natural consistency conditions: (See Lamperti (1977).)

Theorem 2.1.5 (Kolmogorov's extension theorem).

For all $t_1, \dots, t_k \in T$, $k \in \mathbf{N}$ let $\nu_{t_1, \dots, t_k}$ be probability measures on $\mathbf{R}^{nk}$ s.t.

$$\nu_{t_{\sigma(1)}, \dots, t_{\sigma(k)}}(F_1 \times \dots \times F_k) = \nu_{t_1, \dots, t_k}(F_{\sigma^{-1}(1)} \times \dots \times F_{\sigma^{-1}(k)}) \quad (\text{K1})$$

for all permutations σ on $\{1, 2, \dots, k\}$ and

$$\nu_{t_1, \dots, t_k}(F_1 \times \dots \times F_k) = \nu_{t_1, \dots, t_k, t_{k+1}, \dots, t_{k+m}}(F_1 \times \dots \times F_k \times \mathbf{R}^n \times \dots \times \mathbf{R}^n) \quad (\text{K2})$$

for all $m \in \mathbf{N}$, where (of course) the set on the right hand side has a total of $k + m$ factors.

Then there exists a probability space $(\Omega, \mathcal{F}, P)$ and a stochastic process $\{X_t\}$ on Ω , $X_t: \Omega \rightarrow \mathbf{R}^n$, s.t.

$$\nu_{t_1, \dots, t_k}(F_1 \times \dots \times F_k) = P[X_{t_1} \in F_1, \dots, X_{t_k} \in F_k],$$

for all $t_i \in T$, $k \in \mathbf{N}$ and all Borel sets F_i .

2.2 An Important Example: Brownian Motion

In 1828 the Scottish botanist Robert Brown observed that pollen grains suspended in liquid performed an irregular motion. The motion was later explained by the random collisions with the molecules of the liquid. To describe the motion mathematically it is natural to use the concept of a stochastic process $B_t(\omega)$, interpreted as the position at time t of the pollen grain ω . We will generalize slightly and consider an n -dimensional analog.

To construct $\{B_t\}_{t \geq 0}$ it suffices, by the Kolmogorov extension theorem, to specify a family $\{\nu_{t_1, \dots, t_k}\}$ of probability measures satisfying (K1) and (K2). These measures will be chosen so that they agree with our observations of the pollen grain behaviour:

Fix $x \in \mathbf{R}^n$ and define

$$p(t, x, y) = (2\pi t)^{-n/2} \cdot \exp\left(-\frac{|x - y|^2}{2t}\right) \quad \text{for } y \in \mathbf{R}^n, t > 0.$$

If $0 \leq t_1 \leq t_2 \leq \dots \leq t_k$ define a measure $\nu_{t_1, \dots, t_k}$ on $\mathbf{R}^{nk}$ by

$$\begin{aligned} \nu_{t_1, \dots, t_k}(F_1 \times \dots \times F_k) &= \\ &= \int_{F_1 \times \dots \times F_k} p(t_1, x, x_1) p(t_2 - t_1, x_1, x_2) \dots p(t_k - t_{k-1}, x_{k-1}, x_k) dx_1 \dots dx_k \end{aligned} \quad (2.2.1)$$

where we use the notation $dy = dy_1 \dots dy_n$ for Lebesgue measure and the convention that $p(0, x, y) dy = \delta_x(y)$, the unit point mass at x .

Extend this definition to all finite sequences of t_i 's by using (K1). Since $\int_{\mathbf{R}^n} p(t, x, y) dy = 1$ for all $t \geq 0$, (K2) holds, so by Kolmogorov's theorem

there exists a probability space $(\Omega, \mathcal{F}, P^x)$ and a stochastic process $\{B_t\}_{t \geq 0}$ on Ω such that the finite-dimensional distributions of B_t are given by (2.2.1), i.e.

$$\begin{aligned} & P^x(B_{t_1} \in F_1, \dots, B_{t_k} \in F_k) = \\ &= \int_{F_1 \times \dots \times F_k} p(t_1, x, x_1) \cdots p(t_k - t_{k-1}, x_{k-1}, x_k) dx_1 \dots dx_k. \end{aligned} \quad (2.2.2)$$

Definition 2.2.1. *Such a process is called (a version of) Brownian motion starting at x (observe that $P^x(B_0 = x) = 1$).*

The Brownian motion thus defined is not unique, i.e. there exist several quadruples $(B_t, \Omega, \mathcal{F}, P^x)$ such that (2.2.2) holds. However, for our purposes this is not important, we may simply choose any version to work with. As we shall soon see, the paths of a Brownian motion are (or, more correctly, can be chosen to be) continuous, a.s. Therefore we may identify (a.a.) $\omega \in \Omega$ with a continuous function $t \rightarrow B_t(\omega)$ from $[0, \infty)$ into $\mathbf{R}^n$. Thus we may adopt the point of view that Brownian motion is just the space $C([0, \infty), \mathbf{R}^n)$ equipped with certain probability measures P^x (given by (2.2.1) and (2.2.2) above). This version is called the *canonical* Brownian motion. Besides having the advantage of being intuitive, this point of view is useful for the further analysis of measures on $C([0, \infty), \mathbf{R}^n)$, since this space is Polish (i.e. a complete separable metric space). See Stroock and Varadhan (1979).

We state some basic properties of Brownian motion:

- (i) B_t is a *Gaussian process*, i.e. for all $0 \leq t_1 \leq \dots \leq t_k$ the random variable $Z = (B_{t_1}, \dots, B_{t_k}) \in \mathbf{R}^{nk}$ has a *(multi)normal distribution*. This means that there exists a vector $M \in \mathbf{R}^{nk}$ and a non-negative definite matrix $C = [c_{jm}] \in \mathbf{R}^{nk \times nk}$ (the set of all $nk \times nk$ -matrices with real entries) such that

$$E^x \left[\exp \left(i \sum_{j=1}^{nk} u_j Z_j \right) \right] = \exp \left(-\frac{1}{2} \sum_{j,m} u_j c_{jm} u_m + i \sum_j u_j M_j \right) \quad (2.2.3)$$

for all $u = (u_1, \dots, u_{nk}) \in \mathbf{R}^{nk}$, where $i = \sqrt{-1}$ is the imaginary unit and E^x denotes expectation with respect to P^x . Moreover, if (2.2.3) holds then

$$M = E^x[Z] \quad \text{is the mean value of } Z \quad (2.2.4)$$

and

$$c_{jm} = E^x[(Z_j - M_j)(Z_m - M_m)] \quad \text{is the covariance matrix of } Z. \quad (2.2.5)$$

(See Appendix A).

To see that (2.2.3) holds for $Z = (B_{t_1}, \dots, B_{t_k})$ we calculate its left hand side explicitly by using (2.2.2) (see Appendix A) and obtain (2.2.3) with

$$M = E^x[Z] = (x, x, \dots, x) \in \mathbf{R}^{nk} \quad (2.2.6)$$

and

$$C = \begin{pmatrix} t_1 I_n & t_1 I_n & \cdots & t_1 I_n \\ t_1 I_n & t_2 I_n & \cdots & t_2 I_n \\ \vdots & \vdots & \ddots & \vdots \\ t_1 I_n & t_2 I_n & \cdots & t_k I_n \end{pmatrix}. \quad (2.2.7)$$

Hence

$$E^x[B_t] = x \quad \text{for all } t \geq 0 \quad (2.2.8)$$

and

$$E^x[(B_t - x)^2] = nt, E^x[(B_t - x)(B_s - x)] = n \min(s, t). \quad (2.2.9)$$

Moreover,

$$E^x[(B_t - B_s)^2] = n(t - s) \text{ if } t \geq s, \quad (2.2.10)$$

since

$$\begin{aligned} E^x[(B_t - B_s)^2] &= E^x[(B_t - x)^2 - 2(B_t - x)(B_s - x) + (B_s - x)^2] \\ &= n(t - 2s + s) = n(t - s), \text{ when } t \geq s. \end{aligned}$$

(ii) B_t has *independent increments*, i.e.

$$\begin{aligned} B_{t_1}, B_{t_2} - B_{t_1}, \dots, B_{t_k} - B_{t_{k-1}} &\text{ are independent} \\ \text{for all } 0 \leq t_1 < t_2 < \dots < t_k. \end{aligned} \quad (2.2.11)$$

To prove this we use the fact that normal random variables are independent iff they are uncorrelated. (See Appendix A). So it is enough to prove that

$$E^x[(B_{t_i} - B_{t_{i-1}})(B_{t_j} - B_{t_{j-1}})] = 0 \quad \text{when } t_i < t_j, \quad (2.2.12)$$

which follows from the form of C :

$$\begin{aligned} E^x[B_{t_i} B_{t_j} - B_{t_{i-1}} B_{t_j} - B_{t_i} B_{t_{j-1}} + B_{t_{i-1}} B_{t_{j-1}}] \\ = n(t_i - t_{i-1} - t_i + t_{i-1}) = 0. \end{aligned}$$

From this we deduce that $B_s - B_t$ is independent of $\mathcal{F}_t$ if $s > t$.

- (iii) Finally we ask: Is $t \rightarrow B_t(\omega)$ continuous for almost all ω ? Stated like this the question does not make sense, because the set $H = \{\omega; t \rightarrow B_t(\omega) \text{ is continuous}\}$ is not measurable with respect to the Borel σ -algebra $\mathcal{B}$ on $(\mathbf{R}^n)^{[0, \infty)}$ mentioned above (H involves an uncountable number of t 's). However, if modified slightly the question can be given a positive answer. To explain this we need the following important concept:

Definition 2.2.2. Suppose that $\{X_t\}$ and $\{Y_t\}$ are stochastic processes on $(\Omega, \mathcal{F}, P)$. Then we say that $\{X_t\}$ is a version of (or a modification of) $\{Y_t\}$ if

$$P(\{\omega; X_t(\omega) = Y_t(\omega)\}) = 1 \quad \text{for all } t.$$

Note that if X_t is a version of Y_t , then X_t and Y_t have the same finite-dimensional distributions. Thus from the point of view that a stochastic process is a probability law on $(\mathbf{R}^n)^{[0, \infty)}$ two such processes are the same, but nevertheless their path properties may be different. (See Exercise 2.9.)

The continuity question of Brownian motion can be answered by using another famous theorem of Kolmogorov:

Theorem 2.2.3 (Kolmogorov's continuity theorem). Suppose that the process $X = \{X_t\}_{t \geq 0}$ satisfies the following condition: For all $T > 0$ there exist positive constants α, β, D such that

$$E[|X_t - X_s|^\alpha] \leq D \cdot |t - s|^{1+\beta}; \quad 0 \leq s, t \leq T. \quad (2.2.13)$$

Then there exists a continuous version of X .

For a proof see for example Stroock and Varadhan (1979, p. 51).

For Brownian motion B_t it is not hard to prove that (See Exercise 2.8)

$$E^x[|B_t - B_s|^4] = n(n+2)|t - s|^2. \quad (2.2.14)$$

So Brownian motion satisfies Kolmogorov's condition (2.2.13) with $\alpha = 4$, $D = n(n+2)$ and $\beta = 1$, and therefore it has a continuous version. From now on we will assume that B_t is such a continuous version.

Finally we note that

If $B_t = (B_t^{(1)}, \dots, B_t^{(n)})$ is n -dimensional Brownian motion, then the 1-dimensional processes $\{B_t^{(j)}\}_{t \geq 0}$, $1 \leq j \leq n$ are independent, 1-dimensional Brownian motions. (2.2.15)

Exercises

2.1. Suppose that $X: \Omega \rightarrow \mathbf{R}$ is a function which assumes only countably many values $a_1, a_2, \dots \in \mathbf{R}$.

a) Show that X is a random variable if and only if

$$X^{-1}(a_k) \in \mathcal{F} \quad \text{for all } k = 1, 2, \dots \quad (2.2.16)$$

b) Suppose (2.2.16) holds. Show that

$$E[|X|] = \sum_{k=1}^{\infty} |a_k| P[X = a_k]. \quad (2.2.17)$$

- c) If (2.2.16) holds and $E[|X|] < \infty$, show that

$$E[X] = \sum_{k=1}^{\infty} a_k P[X = a_k] .$$

- d) If (2.2.16) holds and $f: \mathbf{R} \rightarrow \mathbf{R}$ is measurable and bounded, show that

$$E[f(X)] = \sum_{k=1}^{\infty} f(a_k) P[X = a_k] .$$

- 2.2.** Let $X: \Omega \rightarrow \mathbf{R}$ be a random variable. The *distribution function* F of X is defined by

$$F(x) = P[X \leq x] .$$

- a) Prove that F has the following properties:

- (i) $0 \leq F \leq 1$, $\lim_{x \rightarrow -\infty} F(x) = 0$, $\lim_{x \rightarrow \infty} F(x) = 1$.
- (ii) F is increasing (= non-decreasing).
- (iii) F is right-continuous, i.e. $F(x) = \lim_{h \rightarrow 0, h > 0} F(x+h)$.

- b) Let $g: \mathbf{R} \rightarrow \mathbf{R}$ be measurable such that $E[|g(X)|] < \infty$. Prove that

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) dF(x) ,$$

where the integral on the right is interpreted in the Lebesgue-Stieltjes sense.

- c) Let $p(x) \geq 0$ be a measurable function on $\mathbf{R}$. We say that X has the *density* p if

$$F(x) = \int_{-\infty}^x p(y) dy \quad \text{for all } x .$$

Thus from (2.2.1)–(2.2.2) we know that 1-dimensional Brownian motion B_t at time t with $B_0 = 0$ has the density

$$p(x) = \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{x^2}{2t}\right); \quad x \in \mathbf{R} .$$

Find the density of B_t^2 .

- 2.3.** Let $\{\mathcal{H}_i\}_{i \in I}$ be a family of σ -algebras on Ω . Prove that

$$\mathcal{H} = \bigcap \{\mathcal{H}_i; i \in I\}$$

is again a σ -algebra.

2.4. a) Let $X: \Omega \rightarrow \mathbf{R}^n$ be a random variable such that

$$E[|X|^p] < \infty \quad \text{for some } p, \quad 0 < p < \infty .$$

Prove *Chebyshev's inequality*:

$$P[|X| \geq \lambda] \leq \frac{1}{\lambda^p} E[|X|^p] \quad \text{for all } \lambda \geq 0 .$$

Hint: $\int_{\Omega} |X|^p dP \geq \int_A |X|^p dP$, where $A = \{\omega: |X| \geq \lambda\}$.

b) Suppose there exists $k > 0$ such that

$$M = E[\exp(k|X|)] < \infty .$$

Prove that $P[|X| \geq \lambda] \leq Me^{-k\lambda}$ for all $\lambda \geq 0$.

2.5. Let $X, Y: \Omega \rightarrow \mathbf{R}$ be two independent random variables and assume for simplicity that X and Y are bounded. Prove that

$$E[XY] = E[X]E[Y] .$$

(Hint: Assume $|X| \leq M$, $|Y| \leq N$. Approximate X and Y by simple functions $\varphi(\omega) = \sum_{i=1}^m a_i \chi_{F_i}(\omega)$, $\psi(\omega) = \sum_{j=1}^n b_j \chi_{G_j}(\omega)$, respectively, where $F_i = X^{-1}([a_i, a_{i+1}))$, $G_j = Y^{-1}([b_j, b_{j+1}))$, $-M = a_0 < a_1 < \dots < a_m = M$, $-N = b_0 < b_1 < \dots < b_n = N$. Then

$$E[X] \approx E[\varphi] = \sum_i a_i P(F_i), \quad E[Y] \approx E[\psi] = \sum_j b_j P(G_j)$$

and

$$E[XY] \approx E[\varphi\psi] = \sum_{i,j} a_i b_j P(F_i \cap G_j) \dots$$

2.6. Let $(\Omega, \mathcal{F}, P)$ be a probability space and let $A_1, A_2, \dots$ be sets in $\mathcal{F}$ such that

$$\sum_{k=1}^{\infty} P(A_k) < \infty .$$

Prove the *Borel-Cantelli lemma*:

$$P\left(\bigcap_{m=1}^{\infty} \bigcup_{k=m}^{\infty} A_k\right) = 0 ,$$

i.e. the probability that ω belongs to infinitely many A_k 's is zero.

- 2.7.** a) Suppose $G_1, G_2, \dots, G_n$ are disjoint subsets of Ω such that

$$\Omega = \bigcup_{i=1}^n G_i .$$

Prove that the family $\mathcal{G}$ consisting of $\emptyset$ and all unions of some (or all) of $G_1, \dots, G_n$ constitutes a σ -algebra on Ω .

- b) Prove that any *finite* σ -algebra $\mathcal{F}$ on Ω is of the type described in a).
 c) Let $\mathcal{F}$ be a *finite* σ -algebra on Ω and let $X: \Omega \rightarrow \mathbf{R}$ be $\mathcal{F}$ -measurable. Prove that X assumes only finitely many possible values. More precisely, there exists a disjoint family of subsets $F_1, \dots, F_m \in \mathcal{F}$ and real numbers $c_1, \dots, c_m$ such that

$$X(\omega) = \sum_{i=1}^m c_i \chi_{F_i}(\omega) .$$

- 2.8.** Let B_t be Brownian motion on $\mathbf{R}$, $B_0 = 0$. Put $E = E^0$.

- a) Use (2.2.3) to prove that

$$E[e^{iuB_t}] = \exp\left(-\frac{1}{2}u^2t\right) \quad \text{for all } u \in \mathbf{R} .$$

- b) Use the power series expansion of the exponential function on both sides, compare the terms with the same power of u and deduce that

$$E[B_t^4] = 3t^2$$

and more generally that

$$E[B_t^{2k}] = \frac{(2k)!}{2^k \cdot k!} t^k ; \quad k \in \mathbf{N} .$$

- c) If you feel uneasy about the lack of rigour in the method in b), you can proceed as follows: Prove that (2.2.2) implies that

$$E[f(B_t)] = \frac{1}{\sqrt{2\pi t}} \int_{\mathbf{R}} f(x) e^{-\frac{x^2}{2t}} dx$$

for all functions f such that the integral on the right converges. Then apply this to $f(x) = x^{2k}$ and use integration by parts and induction on k .

- d) Prove (2.2.14), for example by using b) and induction on n .

- 2.9.** To illustrate that the (finite-dimensional) distributions alone do not give all the information regarding the continuity properties of a process, consider the following example:

Let $(\Omega, \mathcal{F}, P) = ([0, \infty), \mathcal{B}, \mu)$ where $\mathcal{B}$ denotes the Borel σ -algebra on $[0, \infty)$ and μ is a probability measure on $[0, \infty)$ with no mass on single points. Define

$$X_t(\omega) = \begin{cases} 1 & \text{if } t = \omega \\ 0 & \text{otherwise} \end{cases}$$

and

$$Y_t(\omega) = 0 \quad \text{for all } (t, \omega) \in [0, \infty) \times [0, \infty) .$$

Prove that $\{X_t\}$ and $\{Y_t\}$ have the same distributions and that X_t is a version of Y_t . And yet we have that $t \rightarrow Y_t(\omega)$ is continuous for all ω , while $t \rightarrow X_t(\omega)$ is discontinuous for all ω .

2.10. A stochastic process X_t is called *stationary* if $\{X_t\}$ has the same distribution as $\{X_{t+h}\}$ for any $h > 0$. Prove that Brownian motion B_t has stationary increments, i.e. that the process $\{B_{t+h} - B_t\}_{h \geq 0}$ has the same distribution for all t .

2.11. Prove (2.2.15).

2.12. Let B_t be Brownian motion and fix $t_0 \geq 0$. Prove that

$$\tilde{B}_t := B_{t_0+t} - B_{t_0} ; \quad t \geq 0$$

is a Brownian motion.

2.13. Let B_t be 2-dimensional Brownian motion and put

$$D_\rho = \{x \in \mathbf{R}^2; |x| < \rho\} \quad \text{for } \rho > 0 .$$

Compute

$$P^0[B_t \in D_\rho] .$$

2.14. Let B_t be n -dimensional Brownian motion and let $K \subset \mathbf{R}^n$ have zero n -dimensional Lebesgue measure. Prove that the expected total length of time that B_t spends in K is zero. (This implies that the *Green measure* associated with B_t is absolutely continuous with respect to Lebesgue measure. See Chapter 9).

2.15. Let B_t be n -dimensional Brownian motion starting at 0 and let $U \in \mathbf{R}^{n \times n}$ be a (constant) orthogonal matrix, i.e. $UU^T = I$. Prove that

$$\tilde{B}_t := UB_t$$

is also a Brownian motion.

2.16. (Brownian scaling). Let B_t be a 1-dimensional Brownian motion and let $c > 0$ be a constant. Prove that

$$\hat{B}_t := \frac{1}{c} B_{c^2 t}$$

is also a Brownian motion.

2.17. If $X_t(\cdot): \Omega \rightarrow \mathbf{R}$ is a continuous stochastic process, then for $p > 0$ the p 'th variation process of X_t , $\langle X, X \rangle_t^{(p)}$ is defined by

$$\langle X, X \rangle_t^{(p)}(\omega) = \lim_{\Delta t_k \rightarrow 0} \sum_{t_k \leq t} |X_{t_{k+1}}(\omega) - X_{t_k}(\omega)|^p \quad (\text{limit in probability})$$

where $0 = t_1 < t_2 < \dots < t_n = t$ and $\Delta t_k = t_{k+1} - t_k$. In particular, if $p = 1$ this process is called the *total variation process* and if $p = 2$ this is called the *quadratic variation process*. (See Exercise 4.7.) For Brownian motion $B_t \in \mathbf{R}$ we now show that the quadratic variation process is simply

$$\langle B, B \rangle_t(\omega) = \langle B, B \rangle_t^{(2)}(\omega) = t \quad \text{a.s.}$$

Proceed as follows:

a) Define

$$\Delta B_k = B_{t_{k+1}} - B_{t_k}$$

and put

$$Y(t, \omega) = \sum_{t_k \leq t} (\Delta B_k(\omega))^2,$$

Show that

$$E[(\sum_{t_k \leq t} (\Delta B_k)^2 - t)^2] = 2 \sum_{t_k \leq t} (\Delta t_k)^2$$

and deduce that $Y(t, \cdot) \rightarrow t$ in $L^2(P)$ as $\Delta t_k \rightarrow \infty$.

b) Use a) to prove that a.a. paths of Brownian motion do not have a bounded variation on $[0, t]$, i.e. the total variation of Brownian motion is infinite, a.s.

3. Itô Integrals

3.1 Construction of the Itô Integral

We now turn to the question of finding a reasonable mathematical interpretation of the “noise” term in the equation of Problem 1 in the Introduction:

$$\frac{dN}{dt} = (r(t) + \text{“noise”})N(t)$$

or more generally in equations of the form

$$\frac{dX}{dt} = b(t, X_t) + \sigma(t, X_t) \cdot \text{“noise”} , \quad (3.1.1)$$

where b and σ are some given functions. Let us first concentrate on the case when the noise is 1-dimensional. It is reasonable to look for some stochastic process W_t to represent the noise term, so that

$$\frac{dX}{dt} = b(t, X_t) + \sigma(t, X_t) \cdot W_t . \quad (3.1.2)$$

Based on many situations, for example in engineering, one is led to assume that W_t has, at least approximately, these properties:

- (i) $t_1 \neq t_2 \Rightarrow W_{t_1}$ and W_{t_2} are independent.
- (ii) $\{W_t\}$ is stationary, i.e. the (joint) distribution of $\{W_{t_1+t}, \dots, W_{t_k+t}\}$ does not depend on t .
- (iii) $E[W_t] = 0$ for all t .

However, it turns out there does *not* exist any “reasonable” stochastic process satisfying (i) and (ii): Such a W_t cannot have continuous paths. (See Exercise 3.11.) If we require $E[W_t^2] = t$ then the function $(t, \omega) \rightarrow W_t(\omega)$ cannot even be measurable, with respect to the σ -algebra $\mathcal{B} \times \mathcal{F}$, where $\mathcal{B}$ is the Borel σ -algebra on $[0, \infty]$. (See Kallianpur (1980, p. 10).)

Nevertheless it is possible to represent W_t as a generalized stochastic process called the *white noise process*.

That the process is *generalized* means that it can be constructed as a probability measure on the space $\mathcal{S}'$ of tempered distributions on $[0, \infty)$, and not as a probability measure on the much smaller space $\mathbf{R}^{[0, \infty)}$, like an

ordinary process can. See e.g. Hida (1980), Adler (1981), Rozanov (1982), Hida, Kuo, Potthoff and Streit (1993) or Holden, Øksendal, Ubøe and Zhang (1996).

We will avoid this kind of construction and rather try to rewrite equation (3.1.2) in a form that suggests a replacement of W_t by a proper stochastic process: Let $0 = t_0 < t_1 < \dots < t_m = t$ and consider a discrete version of (3.1.2):

$$X_{k+1} - X_k = b(t_k, X_k)\Delta t_k + \sigma(t_k, X_k)W_k\Delta t_k, \quad (3.1.3)$$

where

$$X_j = X(t_j), \quad W_k = W_{t_k}, \quad \Delta t_k = t_{k+1} - t_k.$$

We abandon the W_k -notation and replace $W_k\Delta t_k$ by $\Delta V_k = V_{t_{k+1}} - V_{t_k}$, where $\{V_t\}_{t \geq 0}$ is some suitable stochastic process. The assumptions (i), (ii) and (iii) on W_t suggest that V_t should have *stationary independent increments with mean 0*. It turns out that the only such process with continuous paths is the Brownian motion B_t . (See Knight (1981)). Thus we put $V_t = B_t$ and obtain from (3.1.3):

$$X_k = X_0 + \sum_{j=0}^{k-1} b(t_j, X_j)\Delta t_j + \sum_{j=0}^{k-1} \sigma(t_j, X_j)\Delta B_j. \quad (3.1.4)$$

Is it possible to prove that the limit of the right hand side of (3.1.4) exists, in some sense, when $\Delta t_j \rightarrow 0$? If so, then by applying the usual integration notation we should obtain

$$X_t = X_0 + \int_0^t b(s, X_s)ds + \text{“} \int_0^t \sigma(s, X_s)dB_s \text{”} \quad (3.1.5)$$

and we would adopt as a convention that (3.1.2) really means that $X_t = X_t(\omega)$ is a stochastic process satisfying (3.1.5).

Thus, in the remainder of this chapter we will prove the existence, in a certain sense, of

$$\text{“} \int_0^t f(s, \omega)dB_s(\omega) \text{”}$$

where $B_t(\omega)$ is 1-dimensional Brownian motion starting at the origin, for a wide class of functions $f: [0, \infty] \times \Omega \rightarrow \mathbf{R}$. Then, in Chapter 5, we will return to the solution of (3.1.5).

Suppose $0 \leq S < T$ and $f(t, \omega)$ is given. We want to define

$$\int_S^T f(t, \omega)dB_t(\omega). \quad (3.1.6)$$

It is reasonable to start with a definition for a simple class of functions f and then extend by some approximation procedure. Thus, let us first assume that f has the form

$$\phi(t, \omega) = \sum_{j \geq 0} e_j(\omega) \cdot \mathcal{X}_{[j \cdot 2^{-n}, (j+1)2^{-n})}(t) , \quad (3.1.7)$$

where $\mathcal{X}$ denotes the characteristic (indicator) function and n is a natural number. For such functions it is reasonable to define

$$\int_S^T \phi(t, \omega) dB_t(\omega) = \sum_{j \geq 0} e_j(\omega) [B_{t_{j+1}} - B_{t_j}](\omega) , \quad (3.1.8)$$

where

$$t_k = t_k^{(n)} = \begin{cases} k \cdot 2^{-n} & \text{if } S \leq k \cdot 2^{-n} \leq T \\ S & \text{if } k \cdot 2^{-n} < S \\ T & \text{if } k \cdot 2^{-n} > T \end{cases}$$

However, without any further assumptions on the functions $e_j(\omega)$ this leads to difficulties, as the next example shows.

Here – and in the following – E means the same as E^0 , the expectation w.r.t. the law P^0 for Brownian motion starting at 0. And P means the same as P^0 .

Example 3.1.1. Choose

$$\begin{aligned} \phi_1(t, \omega) &= \sum_{j \geq 0} B_{j \cdot 2^{-n}}(\omega) \cdot \mathcal{X}_{[j \cdot 2^{-n}, (j+1)2^{-n})}(t) \\ \phi_2(t, \omega) &= \sum_{j \geq 0} B_{(j+1)2^{-n}}(\omega) \cdot \mathcal{X}_{[j \cdot 2^{-n}, (j+1)2^{-n})}(t) . \end{aligned}$$

Then

$$E \left[\int_0^T \phi_1(t, \omega) dB_t(\omega) \right] = \sum_{j \geq 0} E[B_{t_j} (B_{t_{j+1}} - B_{t_j})] = 0 ,$$

since $\{B_t\}$ has independent increments. But

$$\begin{aligned} E \left[\int_0^T \phi_2(t, \omega) dB_t(\omega) \right] &= \sum_{j \geq 0} E[B_{t_{j+1}} \cdot (B_{t_{j+1}} - B_{t_j})] \\ &= \sum_{j \geq 0} E[(B_{t_{j+1}} - B_{t_j})^2] = T , \quad \text{by (2.2.10)} . \end{aligned}$$

So, in spite of the fact that both ϕ_1 and ϕ_2 appear to be very reasonable approximations to

$$f(t, \omega) = B_t(\omega) ,$$

their integrals according to (3.1.8) are not close to each other at all, no matter how large n is chosen.

This only reflects the fact that the variations of the paths of B_t are too big to enable us to define the integral (3.1.6) in the Riemann-Stieltjes sense. In fact, one can show that the paths $t \rightarrow B_t$ of Brownian motion are nowhere differentiable, almost surely (a.s.). (See Breiman (1968)). In particular, the total variation of the path is infinite, a.s.

In general it is natural to approximate a given function $f(t, \omega)$ by

$$\sum_j f(t_j^*, \omega) \cdot \mathcal{X}_{[t_j, t_{j+1})}(t)$$

where the points t_j^* belong to the intervals $[t_j, t_{j+1}]$, and then define

$\int_S^T f(t, \omega) dB_t(\omega)$ as the limit (in a sense that we will explain) of $\sum_j f(t_j^*, \omega) [B_{t_{j+1}} - B_{t_j}](\omega)$ as $n \rightarrow \infty$. However, the example above shows that – unlike the Riemann-Stieltjes integral – it does make a difference here what points t_j^* we choose. The following two choices have turned out to be the most useful ones:

- 1) $t_j^* = t_j$ (the left end point), which leads to the *Itô integral*, from now on denoted by

$$\int_S^T f(t, \omega) dB_t(\omega) ,$$

and

- 2) $t_j^* = (t_j + t_{j+1})/2$ (the mid point), which leads to the *Stratonovich integral*, denoted by

$$\int_S^T f(t, \omega) \circ dB_t(\omega) .$$

(See Protter (1990, Th. V. 5.30)).

In the end of this chapter we will explain why these choices are the best and discuss the relations and distinctions between the corresponding integrals.

In any case one must restrict oneself to a special class of functions $f(t, \omega)$ in (3.1.6), also if they have the particular form (3.1.7), in order to obtain a reasonable definition of the integral. We will here present Itô's choice $t_j^* = t_j$. The approximation procedure indicated above will work out successfully provided that f has the property that each of the functions $\omega \rightarrow f(t_j, \omega)$ only depends on the behaviour of $B_s(\omega)$ up to time t_j . This leads to the following important concepts:

Definition 3.1.2. Let $B_t(\omega)$ be n -dimensional Brownian motion. Then we define $\mathcal{F}_t = \mathcal{F}_t^{(n)}$ to be the σ -algebra generated by the random variables $B_s(\cdot)$; $s \leq t$. In other words, $\mathcal{F}_t$ is the smallest σ -algebra containing all sets of the form

$$\{\omega; B_{t_1}(\omega) \in F_1, \dots, B_{t_k}(\omega) \in F_k\},$$

where $t_j \leq t$ and $F_j \subset \mathbf{R}^n$ are Borel sets, $j \leq k = 1, 2, \dots$ (We assume that all sets of measure zero are included in $\mathcal{F}_t$).

One often thinks of $\mathcal{F}_t$ as “the history of B_s up to time t ”. A function $h(\omega)$ will be $\mathcal{F}_t$ -measurable if and only if h can be written as the pointwise a.e. limit of sums of functions of the form

$$g_1(B_{t_1})g_2(B_{t_2}) \cdots g_k(B_{t_k}),$$

where $g_1, \dots, g_k$ are bounded continuous functions and $t_j \leq t$ for $j \leq k$, $k = 1, 2, \dots$ (See Exercise 3.14.) Intuitively, that h is $\mathcal{F}_t$ -measurable means that the value of $h(\omega)$ can be decided from the values of $B_s(\omega)$ for $s \leq t$. For example, $h_1(\omega) = B_{t/2}(\omega)$ is $\mathcal{F}_t$ -measurable, while $h_2(\omega) = B_{2t}(\omega)$ is not.

Note that $\mathcal{F}_s \subset \mathcal{F}_t$ for $s < t$ (i.e. $\{\mathcal{F}_t\}$ is *increasing*) and that $\mathcal{F}_t \subset \mathcal{F}$ for all t .

Definition 3.1.3. Let $\{\mathcal{N}_t\}_{t \geq 0}$ be an increasing family of σ -algebras of subsets of Ω . A process $g(t, \omega): [0, \infty) \times \Omega \rightarrow \mathbf{R}^n$ is called $\mathcal{N}_t$ -adapted if for each $t \geq 0$ the function

$$\omega \rightarrow g(t, \omega)$$

is $\mathcal{N}_t$ -measurable.

Thus the process $h_1(t, \omega) = B_{t/2}(\omega)$ is $\mathcal{F}_t$ -adapted, while $h_2(t, \omega) = B_{2t}(\omega)$ is not.

We now describe our class of functions for which the Itô integral will be defined:

Definition 3.1.4. Let $\mathcal{V} = \mathcal{V}(S, T)$ be the class of functions

$$f(t, \omega): [0, \infty) \times \Omega \rightarrow \mathbf{R}$$

such that

- (i) $(t, \omega) \rightarrow f(t, \omega)$ is $\mathcal{B} \times \mathcal{F}$ -measurable, where $\mathcal{B}$ denotes the Borel σ -algebra on $[0, \infty)$.
- (ii) $f(t, \omega)$ is $\mathcal{F}_t$ -adapted.
- (iii) $E \left[\int_S^T f(t, \omega)^2 dt \right] < \infty$.

The Itô Integral

For functions $f \in \mathcal{V}$ we will now show how to define the *Itô integral*

$$\mathcal{I}[f](\omega) = \int_S^T f(t, \omega) dB_t(\omega) ,$$

where B_t is 1-dimensional Brownian motion.

The idea is natural: First we define $\mathcal{I}[\phi]$ for a simple class of functions ϕ . Then we show that each $f \in \mathcal{V}$ can be approximated (in an appropriate sense) by such ϕ 's and we use this to define $\int f dB$ as the limit of $\int \phi dB$ as $\phi \rightarrow f$.

We now give the details of this construction: A function $\phi \in \mathcal{V}$ is called *elementary* if it has the form

$$\phi(t, \omega) = \sum_j e_j(\omega) \cdot \mathcal{X}_{[t_j, t_{j+1})}(t) . \quad (3.1.9)$$

Note that since $\phi \in \mathcal{V}$ each function e_j must be $\mathcal{F}_{t_j}$ -measurable. Thus in Example 3.1.1 above the function ϕ_1 is elementary while ϕ_2 is not.

For elementary functions $\phi(t, \omega)$ we define the integral according to (3.1.8), i.e.

$$\int_S^T \phi(t, \omega) dB_t(\omega) = \sum_{j \geq 0} e_j(\omega) [B_{t_{j+1}} - B_{t_j}](\omega) . \quad (3.1.10)$$

Now we make the following important observation:

Lemma 3.1.5 (The Itô isometry). *If $\phi(t, \omega)$ is bounded and elementary then*

$$E \left[\left(\int_S^T \phi(t, \omega) dB_t(\omega) \right)^2 \right] = E \left[\int_S^T \phi(t, \omega)^2 dt \right] . \quad (3.1.11)$$

Proof. Put $\Delta B_j = B_{t_{j+1}} - B_{t_j}$. Then

$$E[e_i e_j \Delta B_i \Delta B_j] = \begin{cases} 0 & \text{if } i \neq j \\ E[e_j^2] \cdot (t_{j+1} - t_j) & \text{if } i = j \end{cases}$$

using that $e_i e_j \Delta B_i$ and ΔB_j are independent if $i < j$. Thus

$$\begin{aligned} E \left[\left(\int_S^T \phi dB \right)^2 \right] &= \sum_{i,j} E[e_i e_j \Delta B_i \Delta B_j] = \sum_j E[e_j^2] \cdot (t_{j+1} - t_j) \\ &= E \left[\int_S^T \phi^2 dt \right] . \end{aligned}$$

The idea is now to use the isometry (3.1.11) to extend the definition from elementary functions to functions in $\mathcal{V}$. We do this in several steps:

Step 1. *Let $g \in \mathcal{V}$ be bounded and $g(\cdot, \omega)$ continuous for each ω . Then there exist elementary functions $\phi_n \in \mathcal{V}$ such that*

$$E \left[\int_S^T (g - \phi_n)^2 dt \right] \rightarrow 0 \quad \text{as } n \rightarrow \infty .$$

Proof. Define $\phi_n(t, \omega) = \sum_j g(t_j, \omega) \cdot \mathcal{X}_{[t_j, t_{j+1})}(t)$. Then ϕ_n is elementary since $g \in \mathcal{V}$, and

$$\int_S^T (g - \phi_n)^2 dt \rightarrow 0 \quad \text{as } n \rightarrow \infty, \text{ for each } \omega ,$$

since $g(\cdot, \omega)$ is continuous for each ω . Hence $E \left[\int_S^T (g - \phi_n)^2 dt \right] \rightarrow 0$ as $n \rightarrow \infty$, by bounded convergence.

Step 2. *Let $h \in \mathcal{V}$ be bounded. Then there exist bounded functions $g_n \in \mathcal{V}$ such that $g_n(\cdot, \omega)$ is continuous for all ω and n , and*

$$E \left[\int_S^T (h - g_n)^2 dt \right] \rightarrow 0 .$$

Proof. Suppose $|h(t, \omega)| \leq M$ for all (t, ω) . For each n let ψ_n be a non-negative, continuous function on $\mathbf{R}$ such that

- (i) $\psi_n(x) = 0$ for $x \leq -\frac{1}{n}$ and $x \geq 0$
- and
- (ii) $\int_{-\infty}^{\infty} \psi_n(x) dx = 1$

Define

$$g_n(t, \omega) = \int_0^t \psi_n(s - t) h(s, \omega) ds .$$

Then $g_n(\cdot, \omega)$ is continuous for each ω and $|g_n(t, \omega)| \leq M$. Since $h \in \mathcal{V}$ we can show that $g_n(t, \cdot)$ is $\mathcal{F}_t$ -measurable for all t . (This is a subtle point; see e.g. Karatzas and Shreve (1991), p. 133 for details.) Moreover,

$$\int_S^T (g_n(s, \omega) - h(s, \omega))^2 ds \rightarrow 0 \quad \text{as } n \rightarrow \infty, \text{ for each } \omega ,$$

since $\{\psi_n\}_n$ constitutes an approximate identity. (See e.g. Hoffman (1962, p. 22).) So by bounded convergence

$$E \left[\int_S^T (h(t, \omega) - g_n(t, \omega))^2 dt \right] \rightarrow 0 \quad \text{as } n \rightarrow \infty ,$$

as asserted.

Step 3. *Let $f \in \mathcal{V}$. Then there exists a sequence $\{h_n\} \subset \mathcal{V}$ such that h_n is bounded for each n and*

$$E \left[\int_S^T (f - h_n)^2 dt \right] \rightarrow 0 \text{ as } n \rightarrow \infty .$$

Proof. Put

$$h_n(t, \omega) = \begin{cases} -n & \text{if } f(t, \omega) < -n \\ f(t, \omega) & \text{if } -n \leq f(t, \omega) \leq n \\ n & \text{if } f(t, \omega) > n . \end{cases}$$

Then the conclusion follows by dominated convergence.

That completes the approximation procedure.

We are now ready to complete the definition of the Itô integral

$$\int_S^T f(t, \omega) dB_t(\omega) \quad \text{for } f \in \mathcal{V} .$$

If $f \in \mathcal{V}$ we choose, by Steps 1-3, elementary functions $\phi_n \in \mathcal{V}$ such that

$$E \left[\int_S^T |f - \phi_n|^2 dt \right] \rightarrow 0 .$$

Then define

$$\mathcal{I}[f](\omega) := \int_S^T f(t, \omega) dB_t(\omega) := \lim_{n \rightarrow \infty} \int_S^T \phi_n(t, \omega) dB_t(\omega) .$$

The limit exists as an element of $L^2(P)$, since $\left\{ \int_S^T \phi_n(t, \omega) dB_t(\omega) \right\}$ forms a Cauchy sequence in $L^2(P)$, by (3.1.11).

We summarize this as follows:

Definition 3.1.6 (The Itô integral). Let $f \in \mathcal{V}(S, T)$. Then the Itô integral of f (from S to T) is defined by

$$\int_S^T f(t, \omega) dB_t(\omega) = \lim_{n \rightarrow \infty} \int_S^T \phi_n(t, \omega) dB_t(\omega) \quad (\text{limit in } L^2(P)) \quad (3.1.12)$$

where $\{\phi_n\}$ is a sequence of elementary functions such that

$$E \left[\int_S^T (f(t, \omega) - \phi_n(t, \omega))^2 dt \right] \rightarrow 0 \quad \text{as } n \rightarrow \infty. \quad (3.1.13)$$

Note that such a sequence $\{\phi_n\}$ satisfying (3.1.13) exists by Steps 1–3 above. Moreover, by (3.1.11) the limit in (3.1.12) exists and does not depend on the actual choice of $\{\phi_n\}$, as long as (3.1.13) holds. Furthermore, from (3.1.11) and (3.1.12) we get the following important

Corollary 3.1.7 (The Itô isometry).

$$E \left[\left(\int_S^T f(t, \omega) dB_t \right)^2 \right] = E \left[\int_S^T f^2(t, \omega) dt \right] \quad \text{for all } f \in \mathcal{V}(S, T). \quad (3.1.14)$$

Corollary 3.1.8. If $f(t, \omega) \in \mathcal{V}(S, T)$ and $f_n(t, \omega) \in \mathcal{V}(S, T)$ for $n = 1, 2, \dots$ and $E \left[\int_S^T (f_n(t, \omega) - f(t, \omega))^2 dt \right] \rightarrow 0$ as $n \rightarrow \infty$, then

$$\int_S^T f_n(t, \omega) dB_t(\omega) \rightarrow \int_S^T f(t, \omega) dB_t(\omega) \quad \text{in } L^2(P) \text{ as } n \rightarrow \infty.$$

We illustrate this integral with an example:

Example 3.1.9. Assume $B_0 = 0$. Then

$$\int_0^t B_s dB_s = \frac{1}{2} B_t^2 - \frac{1}{2} t.$$

Proof. Put $\phi_n(s, \omega) = \sum B_j(\omega) \cdot \mathcal{X}_{[t_j, t_{j+1})}(s)$, where $B_j = B_{t_j}$. Then

$$\begin{aligned} E \left[\int_0^t (\phi_n - B_s)^2 ds \right] &= E \left[\sum_j \int_{t_j}^{t_{j+1}} (B_j - B_s)^2 ds \right] \\ &= \sum_j \int_{t_j}^{t_{j+1}} (s - t_j) ds = \sum_j \frac{1}{2} (t_{j+1} - t_j)^2 \rightarrow 0 \quad \text{as } \Delta t_j \rightarrow 0. \end{aligned}$$

So by Corollary 3.1.8

$$\int_0^t B_s dB_s = \lim_{\Delta t_j \rightarrow 0} \int_0^t \phi_n dB_s = \lim_{\Delta t_j \rightarrow 0} \sum_j B_j \Delta B_j .$$

(See also Exercise 3.13.) Now

$$\begin{aligned} \Delta(B_j^2) &= B_{j+1}^2 - B_j^2 = (B_{j+1} - B_j)^2 + 2B_j(B_{j+1} - B_j) \\ &= (\Delta B_j)^2 + 2B_j \Delta B_j , \end{aligned}$$

and therefore, since $B_0 = 0$,

$$B_t^2 = \sum_j \Delta(B_j^2) = \sum_j (\Delta B_j)^2 + 2 \sum_j B_j \Delta B_j$$

or

$$\sum_j B_j \Delta B_j = \frac{1}{2} B_t^2 - \frac{1}{2} \sum_j (\Delta B_j)^2 .$$

Since $\sum_j (\Delta B_j)^2 \rightarrow t$ in $L^2(P)$ as $\Delta t_j \rightarrow 0$ (Exercise 2.17), the result follows.

The extra term $-\frac{1}{2}t$ shows that the Itô stochastic integral does not behave like ordinary integrals. In the next chapter we will establish the *Itô formula*, which explains the result in this example and which makes it easy to calculate many stochastic integrals.

3.2 Some properties of the Itô integral

First we observe the following:

Theorem 3.2.1. *Let $f, g \in \mathcal{V}(0, T)$ and let $0 \leq S < U < T$. Then*

- (i) $\int_S^T f dB_t = \int_S^U f dB_t + \int_U^T f dB_t$ for a.a. ω
- (ii) $\int_S^T (cf + g) dB_t = c \cdot \int_S^T f dB_t + \int_S^T g dB_t$ (c constant) for a.a. ω
- (iii) $E\left[\int_S^T f dB_t\right] = 0$
- (iv) $\int_S^T f dB_t$ is $\mathcal{F}_T$ -measurable.

Proof. This clearly holds for all elementary functions, so by taking limits we obtain this for all $f, g \in \mathcal{V}(0, T)$.

An important property of the Itô integral is that it is a *martingale*:

Definition 3.2.2. A filtration (on $(\Omega, \mathcal{F})$) is a family $\mathcal{M} = \{\mathcal{M}_t\}_{t \geq 0}$ of σ -algebras $\mathcal{M}_t \subset \mathcal{F}$ such that

$$0 \leq s < t \Rightarrow \mathcal{M}_s \subset \mathcal{M}_t$$

(i.e. $\{\mathcal{M}_t\}$ is increasing). An n -dimensional stochastic process $\{M_t\}_{t \geq 0}$ on $(\Omega, \mathcal{F}, P)$ is called a martingale with respect to a filtration $\{\mathcal{M}_t\}_{t \geq 0}$ (and with respect to P) if

- (i) M_t is $\mathcal{M}_t$ -measurable for all t ,
- (ii) $E[|M_t|] < \infty$ for all t
and
- (iii) $E[M_s | \mathcal{M}_t] = M_t$ for all $s \geq t$.

Here the expectation in (ii) and the conditional expectation in (iii) is taken with respect to $P = P^0$. (See Appendix B for a survey of conditional expectation).

Example 3.2.3. Brownian motion B_t in $\mathbf{R}^n$ is a martingale w.r.t. the σ -algebras $\mathcal{F}_t$ generated by $\{B_s; s \leq t\}$, because

$$\begin{aligned} E[|B_t|]^2 &\leq E[|B_t|^2] = |B_0|^2 + nt \quad \text{and if } s \geq t \text{ then} \\ E[B_s | \mathcal{F}_t] &= E[B_s - B_t + B_t | \mathcal{F}_t] \\ &= E[B_s - B_t | \mathcal{F}_t] + E[B_t | \mathcal{F}_t] = 0 + B_t = B_t. \end{aligned}$$

Here we have used that $E[(B_s - B_t) | \mathcal{F}_t] = E[B_s - B_t] = 0$ since $B_s - B_t$ is independent of $\mathcal{F}_t$ (see (2.2.11) and Theorem B.2.d)) and we have used that $E[B_t | \mathcal{F}_t] = B_t$ since B_t is $\mathcal{F}_t$ -measurable (see Theorem B.2.c)).

For continuous martingales we have the following important inequality due to Doob: (See e.g. Stroock and Varadhan (1979), Theorem 1.2.3 or Revuz and Yor (1991), Theorem II.1.7)

Theorem 3.2.4 (Doob's martingale inequality). If M_t is a martingale such that $t \rightarrow M_t(\omega)$ is continuous a.s., then for all $p \geq 1, T \geq 0$ and all $\lambda > 0$

$$P\left[\sup_{0 \leq t \leq T} |M_t| \geq \lambda\right] \leq \frac{1}{\lambda^p} \cdot E[|M_T|^p].$$

We now use this inequality to prove that the Itô integral

$$\int_0^t f(s, \omega) dB_s$$

can be chosen to depend continuously on t :

Theorem 3.2.5. *Let $f \in \mathcal{V}(0, T)$. Then there exists a t -continuous version of*

$$\int_0^t f(s, \omega) dB_s(\omega) ; \quad 0 \leq t \leq T ,$$

i.e. there exists a t -continuous stochastic process J_t on $(\Omega, \mathcal{F}, P)$ such that

$$P[J_t = \int_0^t f dB] = 1 \quad \text{for all } t, 0 \leq t \leq T . \quad (3.2.1)$$

Proof. Let $\phi_n = \phi_n(t, \omega) = \sum_j e_j^{(n)}(\omega) \mathcal{X}_{[t_j^{(n)}, t_{j+1}^{(n)})}(t)$ be elementary functions such that

$$E \left[\int_0^T (f - \phi_n)^2 dt \right] \rightarrow 0 \quad \text{when } n \rightarrow \infty .$$

Put

$$I_n(t, \omega) = \int_0^t \phi_n(s, \omega) dB_s(\omega)$$

and

$$I_t = I(t, \omega) = \int_0^t f(s, \omega) dB_s(\omega) ; \quad 0 \leq t \leq T .$$

Then $I_n(\cdot, \omega)$ is continuous, for all n . Moreover, $I_n(t, \omega)$ is a martingale with respect to $\mathcal{F}_t$, for all n :

$$\begin{aligned} E[I_n(s, \omega) | \mathcal{F}_t] &= E \left[\left(\int_0^t \phi_n dB + \int_t^s \phi_n dB \right) \middle| \mathcal{F}_t \right] \\ &= \int_0^t \phi_n dB + E \left[\sum_{t \leq t_j^{(n)} \leq t_{j+1}^{(n)} \leq s} e_j^{(n)} \Delta B_j \middle| \mathcal{F}_t \right] \\ &= \int_0^t \phi_n dB + \sum_j E[E[e_j^{(n)} \Delta B_j | \mathcal{F}_{t_j^{(n)}}] | \mathcal{F}_t] \\ &= \int_0^t \phi_n dB + \sum_j E[e_j^{(n)} E[\Delta B_j | \mathcal{F}_{t_j^{(n)}}] | \mathcal{F}_t] \\ &= \int_0^t \phi_n dB = I_n(t, \omega) \end{aligned} \quad (3.2.2)$$

when $t < s$, using Theorem B.3. and Theorem B.2.d).

Hence $I_n - I_m$ is also an $\mathcal{F}_t$ -martingale, so by the martingale inequality (Theorem 3.2.4) it follows that

$$\begin{aligned} P \left[\sup_{0 \leq t \leq T} |I_n(t, \omega) - I_m(t, \omega)| > \epsilon \right] &\leq \frac{1}{\epsilon^2} \cdot E[|I_n(T, \omega) - I_m(T, \omega)|^2] \\ &= \frac{1}{\epsilon^2} E \left[\int_0^T (\phi_n - \phi_m)^2 ds \right] \rightarrow 0 \quad \text{as } m, n \rightarrow \infty. \end{aligned}$$

Hence we may choose a subsequence $n_k \uparrow \infty$ s.t.

$$P \left[\sup_{0 \leq t \leq T} |I_{n_{k+1}}(t, \omega) - I_{n_k}(t, \omega)| > 2^{-k} \right] < 2^{-k}.$$

By the Borel-Cantelli lemma

$$P \left[\sup_{0 \leq t \leq T} |I_{n_{k+1}}(t, \omega) - I_{n_k}(t, \omega)| > 2^{-k} \text{ for infinitely many } k \right] = 0.$$

So for a.a. ω there exists $k_1(\omega)$ such that

$$\sup_{0 \leq t \leq T} |I_{n_{k+1}}(t, \omega) - I_{n_k}(t, \omega)| \leq 2^{-k} \quad \text{for } k \geq k_1(\omega).$$

Therefore $I_{n_k}(t, \omega)$ is uniformly convergent for $t \in [0, T]$, for a.a. ω and so the limit, denoted by $J_t(\omega)$, is t -continuous for $t \in [0, T]$, a.s. Since $I_{n_k}(t, \cdot) \rightarrow I(t, \cdot)$ in $L^2[P]$ for all t , we must have

$$I_t = J_t \text{ a.s. ,} \quad \text{for all } t \in [0, T].$$

That completes the proof. $\square$

From now on we shall always assume that $\int_0^t f(s, \omega) dB_s(\omega)$ means a t -continuous version of the integral.

Corollary 3.2.6. *Let $f(t, \omega) \in \mathcal{V}(0, T)$ for all T . Then*

$$M_t(\omega) = \int_0^t f(s, \omega) dB_s$$

is a martingale w.r.t. $\mathcal{F}_t$ and

$$P \left[\sup_{0 \leq t \leq T} |M_t| \geq \lambda \right] \leq \frac{1}{\lambda^2} \cdot E \left[\int_0^T f(s, \omega)^2 ds \right]; \quad \lambda, T > 0. \quad (3.2.3)$$

Proof. This follows from (3.2.2), the a.s. t -continuity of M_t and the martingale inequality (Theorem 3.2.4), combined with the Itô isometry (3.1.14). $\square$

3.3 Extensions of the Itô integral

The Itô integral $\int f dB$ can be defined for a larger class of integrands f than $\mathcal{V}$. First, the measurability condition (ii) of Definition 3.1.4 can be relaxed to the following:

- (ii)' There exists an increasing family of σ -algebras $\mathcal{H}_t; t \geq 0$ such that
- a) B_t is a martingale with respect to $\mathcal{H}_t$ and
 - b) f_t is $\mathcal{H}_t$ -adapted.

Note that a) implies that $\mathcal{F}_t \subset \mathcal{H}_t$. The essence of this extension is that we can allow f_t to depend on more than $\mathcal{F}_t$ as long as B_t remains a martingale with respect to the “history” of $f_s; s \leq t$. If (ii)' holds, then $E[B_s - B_t | \mathcal{H}_t] = 0$ for all $s > t$ and if we inspect our proofs above, we see that this is sufficient to carry out the construction of the Itô integral as before.

The most important example of a situation where (ii)' applies (and (ii) doesn't) is the following:

Suppose $B_t(\omega) = B_k(t, \omega)$ is the k 'th coordinate of n -dimensional Brownian motion $(B_1, \dots, B_n)$. Let $\mathcal{F}_t^{(n)}$ be the σ -algebra generated by $B_1(s_1, \cdot), \dots, B_n(s_n, \cdot); s_k \leq t$. Then $B_k(t, \omega)$ is a martingale with respect to $\mathcal{F}_t^{(n)}$ because $B_k(s, \cdot) - B_k(t, \cdot)$ is independent of $\mathcal{F}_t^{(n)}$ when $s > t$. Thus we have now defined $\int_0^t f(s, \omega) dB_k(s, \omega)$ for $\mathcal{F}_t^{(n)}$ -adapted integrands $f(t, \omega)$. That includes integrals like

$$\int B_2 dB_1 \quad \text{or} \quad \int \sin(B_1^2 + B_2^2) dB_2$$

involving several components of n -dimensional Brownian motion. (Here we have used the notation $dB_1 = dB_1(t, \omega)$ etc.)

This allows us to define the **multi-dimensional Itô integral** as follows:

Definition 3.3.1. Let $B = (B_1, B_2, \dots, B_n)$ be n -dimensional Brownian motion. Then $\mathcal{V}_{\mathcal{H}}^{m \times n}(S, T)$ denotes the set of $m \times n$ matrices $v = [v_{ij}(t, \omega)]$ where each entry $v_{ij}(t, \omega)$ satisfies (i) and (iii) of Definition 3.1.4 and (ii)' above, with respect to some filtration $\mathcal{H} = \{\mathcal{H}_t\}_{t \geq 0}$.

If $v \in \mathcal{V}_{\mathcal{H}}^{m \times n}(S, T)$ we define, using matrix notation

$$\int_S^T v dB = \int_S^T \begin{pmatrix} v_{11} & \cdots & v_{1n} \\ \vdots & & \vdots \\ v_{m1} & \cdots & v_{mn} \end{pmatrix} \begin{pmatrix} dB_1 \\ \vdots \\ dB_n \end{pmatrix}$$

to be the $m \times 1$ matrix (column vector) whose i 'th component is the following sum of (extended) 1-dimensional Itô integrals:

$$\sum_{j=1}^n \int_S^T v_{ij}(s, \omega) dB_j(s, \omega) .$$

If $\mathcal{H} = \mathcal{F}^{(n)} = \{\mathcal{F}_t^{(n)}\}_{t \geq 0}$ we write $\mathcal{V}^{m \times n}(S, T)$ and if $m = 1$ we write $\mathcal{V}_{\mathcal{H}}^n(S, T)$ (respectively $\mathcal{V}^n(S, T)$) instead of $\mathcal{V}_{\mathcal{H}}^{n \times 1}(S, T)$ (respectively $\mathcal{V}^{n \times 1}(S, T)$). We also put

$$\mathcal{V}^{m \times n} = \mathcal{V}^{m \times n}(0, \infty) = \bigcap_{T > 0} \mathcal{V}^{m \times n}(0, T) .$$

The next extension of the Itô integral consists of weakening condition (iii) of Definition 3.1.4 to

$$(iii)', \quad P \left[\int_S^T f(s, \omega)^2 ds < \infty \right] = 1 .$$

Definition 3.3.2. $\mathcal{W}_{\mathcal{H}}(S, T)$ denotes the class of processes $f(t, \omega) \in \mathbf{R}$ satisfying (i) of Definition 3.1.4 and (ii)', (iii)' above. Similarly to the notation for $\mathcal{V}$ we put $\mathcal{W}_{\mathcal{H}} = \bigcap_{T > 0} \mathcal{W}_{\mathcal{H}}(0, T)$ and in the matrix case we write $\mathcal{W}_{\mathcal{H}}^{m \times n}(S, T)$ etc. If $\mathcal{H} = \mathcal{F}^{(n)}$ we write $\mathcal{W}(S, T)$ instead of $\mathcal{W}_{\mathcal{F}^{(n)}}(S, T)$ etc. If the dimension is clear from the context we sometimes drop the superscript and write $\mathcal{F}$ for $\mathcal{F}^{(n)}$ and so on.

Let B_t denote 1-dimensional Brownian motion. If $f \in \mathcal{W}_{\mathcal{H}}$ one can show that for all t there exist step functions $f_n \in \mathcal{W}_{\mathcal{H}}$ such that $\int_0^t |f_n - f|^2 ds \rightarrow 0$ in probability, i.e. in measure with respect to P . For such a sequence one has that $\int_0^t f_n(s, \omega) dB_s$ converges in probability to some random variable and the limit only depends on f , not on the sequence $\{f_n\}$. Thus we may define

$$\int_0^t f(s, \omega) dB_s(\omega) = \lim_{n \rightarrow \infty} \int_0^t f_n(s, \omega) dB_s(\omega) \quad (\text{limit in probability}) \quad \text{for } f \in \mathcal{W}_{\mathcal{H}} . \quad (3.3.1)$$

As before there exists a t -continuous version of this integral. See Friedman (1975, Chap. 4) or McKean (1969, Chap. 2) for details. Note, however, that this integral is not in general a martingale. See for example Dudley's Theorem (Theorem 12.1.5). It is, however, a *local* martingale. See Karatzas and Shreve (1991), p. 146. See also Exercise 7.12.

A comparison of Itô and Stratonovich integrals

Let us now return to our original question in this chapter: We have argued that the mathematical interpretation of the white noise equation

$$\frac{dX}{dt} = b(t, X_t) + \sigma(t, X_t) \cdot W_t \quad (3.3.2)$$

is that X_t is a solution of the integral equation

$$X_t = X_0 + \int_0^t b(s, X_s) ds + \text{“} \int_0^t \sigma(s, X_s) dB_s \text{”} , \quad (3.3.3)$$

for some suitable interpretation of the last integral in (3.3.3). However, as indicated earlier, the Itô interpretation of an integral of the form

$$\text{“} \int_0^t f(s, \omega) dB_s(\omega) \text{”} \quad (*)$$

is just one of several reasonable choices. For example, the Stratonovich integral is another possibility, leading (in general) to a different result. So the question still remains: *Which interpretation of (*) makes (3.3.3) the “right” mathematical model for the equation (3.3.2)?* Here is an argument that indicates that the *Stratonovich* interpretation in some situations may be the most appropriate: Choose t -continuously differentiable processes $B_t^{(n)}$ such that for a.a. ω

$$B^{(n)}(t, \omega) \rightarrow B(t, \omega) \quad \text{as } n \rightarrow \infty$$

uniformly (in t) in bounded intervals. For each ω let $X_t^{(n)}(\omega)$ be the solution of the corresponding (deterministic) differential equation

$$\frac{dX_t}{dt} = b(t, X_t) + \sigma(t, X_t) \frac{dB_t^{(n)}}{dt} . \quad (3.3.4)$$

Then $X_t^{(n)}(\omega)$ converges to some function $X_t(\omega)$ in the same sense: For a.a. ω we have that $X_t^{(n)}(\omega) \rightarrow X_t(\omega)$ as $n \rightarrow \infty$, uniformly (in t) in bounded intervals.

It turns out (see Wong and Zakai (1969) and Sussman (1978)) that this solution X_t coincides with the solution of (3.3.3) obtained by using *Stratonovich* integrals, i.e.

$$X_t = X_0 + \int_0^t b(s, X_s) ds + \int_0^t \sigma(s, X_s) \circ dB_s . \quad (3.3.5)$$

This implies that X_t is the solution of the following *modified Itô equation*:

$$X_t = X_0 + \int_0^t b(s, X_s) ds + \frac{1}{2} \int_0^t \sigma'(s, X_s) \sigma(s, X_s) ds + \int_0^t \sigma(s, X_s) dB_s , \quad (3.3.6)$$

where σ' denotes the derivative of $\sigma(t, x)$ with respect to x . (See Stratonovich (1966)).

Therefore, from this point of view it seems reasonable to use (3.3.6) (i.e. the Stratonovich interpretation) – and not the Itô interpretation

$$X_t = X_0 + \int_0^t b(s, X_s) ds + \int_0^t \sigma(s, X_s) dB_s \quad (3.3.7)$$

as the model for the original white noise equation (3.3.2).

On the other hand, the specific feature of the Itô model of “not looking into the future” (as explained after Example 3.1.1) seems to be a reason for choosing the Itô interpretation in many cases, for example in biology (see the discussion in Turelli (1977)). The difference between the two interpretations is illustrated in Example 5.1.1. Note that (3.3.6) and (3.3.7) coincide if $\sigma(t, x)$ does not depend on x . For example, this is the situation in the linear case handled in the filtering problem in Chapter 6.

In any case, because of the explicit connection (3.3.6) between the two models (and a similar connection in higher dimensions – see (6.1.3)), it will for many purposes suffice to do the general mathematical treatment for one of the two types of integrals. In general one can say that the Stratonovich integral has the advantage of leading to ordinary chain rule formulas under a transformation (change of variable), i.e. there are no second order terms in the Stratonovich analogue of the Itô transformation formula (see Theorems 4.1.2 and 4.2.1). This property makes the Stratonovich integral natural to use for example in connection with stochastic differential equations on manifolds (see Elworthy (1982) or Ikeda and Watanabe (1989)).

However, Stratonovich integrals are not martingales, as we have seen that Itô integrals are. This gives the Itô integral an important computational advantage, even though it does not behave so nicely under transformations (as Example 3.1.9 shows). For our purposes the Itô integral will be most convenient, so we will base our discussion on that from now on.

Exercises

Unless otherwise stated B_t denotes Brownian motion in $\mathbf{R}$, $B_0 = 0$.

3.1. Prove directly from the definition of Itô integrals (Definition 3.1.6) that

$$\int_0^t s dB_s = tB_t - \int_0^t B_s ds .$$

(Hint: Note that

$$\sum_j \Delta(s_j B_j) = \sum_j s_j \Delta B_j + \sum_j B_{j+1} \Delta s_j .)$$

3.2. Prove directly from the definition of Itô integrals that

$$\int_0^t B_s^2 dB_s = \frac{1}{3} B_t^3 - \int_0^t B_s ds .$$

3.3. If $X_t: \Omega \rightarrow \mathbf{R}^n$ is a stochastic process, let $\mathcal{H}_t = \mathcal{H}_t^{(X)}$ denote the σ -algebra generated by $\{X_s(\cdot); s \leq t\}$ (i.e. $\{\mathcal{H}_t^{(X)}\}_{t \geq 0}$ is the *filtration of the process* $\{X_t\}_{t \geq 0}$).

- a) Show that if X_t is a martingale w.r.t. *some* filtration $\{\mathcal{N}_t\}_{t \geq 0}$, then X_t is also a martingale w.r.t. its own filtration $\{\mathcal{H}_t^{(X)}\}_{t \geq 0}$.
 b) Show that if X_t is a martingale w.r.t. $\mathcal{H}_t^{(X)}$, then

$$E[X_t] = E[X_0] \quad \text{for all } t \geq 0 . \quad (*)$$

- c) Give an example of a stochastic process X_t satisfying $(*)$ and which is *not* a martingale w.r.t. its own filtration.

3.4. Check whether the following processes X_t are martingales w.r.t. $\{\mathcal{F}_t\}$:

- (i) $X_t = B_t + 4t$
 (ii) $X_t = B_t^2$
 (iii) $X_t = t^2 B_t - 2 \int_0^t s B_s ds$
 (iv) $X_t = B_1(t) B_2(t)$, where $(B_1(t), B_2(t))$ is 2-dimensional Brownian motion.

3.5. Prove directly (without using Example 3.1.9) that

$$M_t = B_t^2 - t$$

is an $\mathcal{F}_t$ -martingale.

3.6. Prove that $N_t = B_t^3 - 3tB_t$ is a martingale.

3.7. A famous result of Itô (1951) gives the following formula for n times iterated Itô integrals:

$$n! \int \cdots \left(\int_{0 \leq u_1 \leq \cdots \leq u_n \leq t} dB_{u_1} dB_{u_2} \cdots dB_{u_n} \right) = t^{\frac{n}{2}} h_n \left(\frac{B_t}{\sqrt{t}} \right) \quad (3.3.8)$$

where h_n is the *Hermite polynomial* of degree n , defined by

$$h_n(x) = (-1)^n e^{\frac{x^2}{2}} \frac{d^n}{dx^n} \left(e^{-\frac{x^2}{2}} \right) ; \quad n = 0, 1, 2, \dots$$

(Thus $h_0(x) = 1$, $h_1(x) = x$, $h_2(x) = x^2 - 1$, $h_3(x) = x^3 - 3x$.)

- a) Verify that in each of these n Itô integrals the integrand satisfies the requirements in Definition 3.1.4.

- b) Verify formula (3.3.8) for $n = 1, 2, 3$ by combining Example 3.1.9 and Exercise 3.2.
 c) Use b) to give a new proof of the statement in Exercise 3.6.
3.8. a) Let Y be a real valued random variable on $(\Omega, \mathcal{F}, P)$ such that

$$E[|Y|] < \infty .$$

Define

$$M_t = E[Y|\mathcal{F}_t] ; \quad t \geq 0 .$$

Show that M_t is an $\mathcal{F}_t$ -martingale.

- b) Conversely, let $M_t; t \geq 0$ be a real valued $\mathcal{F}_t$ -martingale such that

$$\sup_{t \geq 0} E[|M_t|^p] < \infty \quad \text{for some } p > 1 .$$

Show that there exists $Y \in L^1(P)$ such that

$$M_t = E[Y|\mathcal{F}_t] .$$

(Hint: Use Corollary C.7.)

- 3.9.** Suppose $f \in \mathcal{V}(0, T)$ and that $t \rightarrow f(t, \omega)$ is continuous for a.a. ω . Then we have shown that

$$\int_0^T f(t, \omega) dB_t(\omega) = \lim_{\Delta t_j \rightarrow 0} \sum_j f(t_j, \omega) \Delta B_j \quad \text{in } L^2(P) .$$

Similarly we define the *Stratonovich integral* of f by

$$\int_0^T f(t, \omega) \circ dB_t(\omega) = \lim_{\Delta t_j \rightarrow 0} \sum_j f(t_j^*, \omega) \Delta B_j , \quad \text{where } t_j^* = \frac{1}{2}(t_j + t_{j+1}) ,$$

whenever the limit exists in $L^2(P)$. In general these integrals are different. For example, compute

$$\int_0^T B_t \circ dB_t$$

and compare with Example 3.1.9.

- 3.10.** If the function f in Exercise 3.9 varies “smoothly” with t then in fact the Itô and Stratonovich integrals of f coincide. More precisely, assume that there exists $K < \infty$ and $\epsilon > 0$ such that

$$E[|f(s, \cdot) - f(t, \cdot)|^2] \leq K|s - t|^{1+\epsilon} ; \quad 0 \leq s, t \leq T .$$

Prove that then we have

$$\int_0^T f(t, \omega) dB_t = \lim_{\Delta t_j \rightarrow 0} \sum_j f(t'_j, \omega) \Delta B_j \quad (\text{limit in } L^1(P))$$

for any choice of $t'_j \in [t_j, t_{j+1}]$. In particular,

$$\int_0^T f(t, \omega) dB_t = \int_0^T f(t, \omega) \circ dB_t .$$

(Hint: Consider $E[\sum_j f(t_j, \omega) \Delta B_j - \sum_j f(t'_j, \omega) \Delta B_j]$.)

- 3.11.** Let W_t be a stochastic process satisfying (i), (ii) and (iii) (below (3.1.2)). Prove that W_t cannot have continuous paths. (Hint: Consider $E[(W_t^{(N)} - W_s^{(N)})^2]$, where

$$W_t^{(N)} = (-N) \vee (N \wedge W_t), \quad N = 1, 2, 3, \dots .$$

- 3.12.** As in Exercise 3.9 we let $\circ dB_t$ denote Stratonovich differentials.

- (i) Use (3.3.6) to transform the following Stratonovich differential equations into Itô differential equations:
 - (a) $dX_t = \gamma X_t dt + \alpha X_t \circ dB_t$
 - (b) $dX_t = \sin X_t \cos X_t dt + (t^2 + \cos X_t) \circ dB_t$
- (ii) Transform the following Itô differential equations into Stratonovich differential equations:
 - (a) $dX_t = r X_t dt + \alpha X_t dB_t$
 - (b) $dX_t = 2e^{-X_t} dt + X_t^2 dB_t$

- 3.13.** A stochastic process $X_t(\cdot): \Omega \rightarrow \mathbf{R}$ is *continuous in mean square* if $E[X_t^2] < \infty$ for all t and

$$\lim_{s \rightarrow t} E[(X_s - X_t)^2] = 0 \quad \text{for all } t \geq 0 .$$

- a) Prove that Brownian motion B_t is continuous in mean square.
- b) Let $f: \mathbf{R} \rightarrow \mathbf{R}$ be a Lipschitz continuous function, i.e. there exists $C < \infty$ such that

$$|f(x) - f(y)| \leq C|x - y| \quad \text{for all } x, y \in \mathbf{R} .$$

Prove that

$$Y_t := f(B_t)$$

is continuous in mean square.

- c) Let X_t be a stochastic process which is continuous in mean square and assume that $X_t \in \mathcal{V}(S, T)$, $T < \infty$. Show that

$$\int_S^T X_t dB_t = \lim_{n \rightarrow \infty} \int_S^T \phi_n(t, \omega) dB_t(\omega) \quad (\text{limit in } L^2(P))$$

where

$$\phi_n(t, \omega) = \sum_j X_{t_j^{(n)}}(\omega) \mathcal{X}_{[t_j^{(n)}, t_{j+1}^{(n)})}(t), \quad T < \infty.$$

(Hint: Consider

$$E \left[\int_S^T (X_t - \phi_n(t))^2 dt \right] = E \left[\sum_j \int_{t_j^{(n)}}^{t_{j+1}^{(n)}} (X_t - X_{t_j^{(n)}})^2 dt \right].$$

- 3.14.** Show that a function $h(\omega)$ is $\mathcal{F}_t$ -measurable if and only if h is a pointwise limit (for a.a. ω) of sums of functions of the form

$$g_1(B_{t_1}) \cdot g_2(B_{t_2}) \cdots g_k(B_{t_k})$$

where $g_1, \dots, g_k$ are bounded continuous functions and $t_j \leq t$ for $j \leq k$, $k = 1, 2, \dots$

Hint: Complete the following steps:

- a) We may assume that h is bounded.
b) For $n = 1, 2, \dots$ and $j = 1, 2, \dots$ put $t_j = t_j^{(n)} = j \cdot 2^{-n}$. For fixed n let $\mathcal{H}_n$ be the σ -algebra generated by $\{B_{t_j}(\cdot)\}_{t_j \leq t}$. Then by Corollary C.9

$$h = E[h|\mathcal{F}_t] = \lim_{n \rightarrow \infty} E[h|\mathcal{H}_n] \quad (\text{pointwise a.e. limit})$$

- c) Define $h_n := E[h|\mathcal{H}_n]$. Then by the Doob-Dynkin lemma (Lemma 2.1.2) we have

$$h_n(\omega) = G_n(B_{t_1}(\omega), \dots, B_{t_k}(\omega))$$

for some Borel function $G_n: \mathbf{R}^k \rightarrow \mathbf{R}$, where $k = \max\{j; j \cdot 2^{-n} \leq t\}$. Now use that any Borel function $G: \mathbf{R}^k \rightarrow \mathbf{R}$ can be approximated pointwise a.e. by a continuous function $F: \mathbf{R}^k \rightarrow \mathbf{R}$ and complete the proof by applying the Stone-Weierstrass theorem.

- 3.15.** Suppose $f, g \in \mathcal{V}(S, T)$ and that there exist constants C, D such that

$$C + \int_S^T f(t, \omega) dB_t(\omega) = D + \int_S^T g(t, \omega) dB_t(\omega) \quad \text{for a.a. } \omega \in \Omega.$$

Show that

$$C = D$$

and

$$f(t, \omega) = g(t, \omega) \quad \text{for a.a. } (t, \omega) \in [S, T] \times \Omega .$$

- 3.16.** Let $X: \Omega \rightarrow \mathbf{R}$ be a random variable such that $E[X^2] < \infty$ and let $\mathcal{H} \subset \mathcal{F}$ be a σ -algebra. Show that

$$E[(E[X|\mathcal{H}])^2] \leq E[X^2] .$$

(See Lemma 6.1.1. See also the Jensen inequality for conditional expectation (Appendix B).)

- 3.17.** Let $(\Omega, \mathcal{F}, P)$ be a probability space and let $X: \Omega \rightarrow \mathbf{R}$ be a random variable with $E[|X|] < \infty$. If $\mathcal{G} \subset \mathcal{F}$ is a *finite* σ -algebra, then by Exercise 2.7 there exists a partition $\Omega = \bigcup_{i=1}^n G_i$ such that $\mathcal{G}$ consists of $\emptyset$ and unions of some (or all) of $G_1, \dots, G_n$.

- a) Explain why $E[X|\mathcal{G}](\omega)$ is constant on each G_i . (See Exercise 2.7 c).)
 b) Assume that $P[G_i] > 0$. Show that

$$E[X|\mathcal{G}](\omega) = \frac{\int_{G_i} X dP}{P(G_i)} \quad \text{for } \omega \in G_i .$$

- c) Suppose X assumes only finitely many values $a_1, \dots, a_m$. Then from elementary probability theory we know that (see Exercise 2.1)

$$E[X|G_i] = \sum_{k=1}^m a_k P[X = a_k | G_i] .$$

Compare with b) and verify that

$$E[X|G_i] = E[X|\mathcal{G}](\omega) \quad \text{for } \omega \in G_i .$$

Thus we may regard the conditional expectation as defined in Appendix B as a (substantial) generalization of the conditional expectation in elementary probability theory.